

Executive Business Report

Project: End-to-End Sales Forecasting & Demand Intelligence System

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1. Executive Summary

This project develops a sales forecasting and demand intelligence system using the Superstore Sales dataset. Historical sales data was analyzed to identify trends, seasonality, demand patterns, and unusual sales behaviour. Three forecasting models (SARIMA, Facebook Prophet, and XGBoost) were evaluated, with XGBoost delivering the best predictive performance. The project also includes anomaly detection, product demand segmentation using K-Means clustering, and an interactive Streamlit dashboard to support inventory and business decision-making.

2. Key Findings

- Technology generated the strongest forecasted growth among the evaluated categories.
- East region showed strong future sales potential.
- Seasonal fluctuations were visible in monthly sales, justifying time-series forecasting.
- XGBoost achieved the lowest forecasting error compared with SARIMA and Prophet.

3. Three-Month Sales Forecast

The forecasting analysis indicates relatively stable sales with stronger expected demand for the Technology category and the East region. Based on the evaluation metrics, XGBoost is recommended for production forecasting because it achieved the lowest MAE, RMSE, and MAPE among the tested models.

4. Anomaly Detection

Sales anomalies were identified using Isolation Forest and Z-Score analysis. The most significant positive anomaly is likely associated with seasonal promotions or festive shopping periods. These anomalies should be investigated before making inventory decisions.

5. Product Demand Segmentation

Segment	Recommended Stocking Strategy
High Volume / Stable Demand	Maintain higher inventory levels and frequent replenishment.
Growing Demand	Increase stock gradually while monitoring monthly demand.

Low Volume / High Volatility	Maintain lean inventory and use targeted promotions.
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6. Business Recommendations

1. Prioritize inventory for Technology products and high-growth regions.
2. Use the XGBoost model for monthly demand planning and replenishment.
3. Monitor detected anomalies to distinguish promotional effects from operational issues.

7. Limitation

Forecasts are based on historical sales patterns and do not explicitly incorporate external factors such as economic conditions, competitor actions, or future marketing campaigns.

Conclusion

The completed system combines forecasting, anomaly detection, clustering, and dashboard visualization into a practical decision-support solution for retail inventory planning.